

2.6 Sequences diverging to $\pm\infty$

Definition 17. Let (x_n) be a real sequence. We say that (x_n) diverges to $+\infty$ (or simply diverges to ∞) if, for every $R \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies x_n > R.$$

In this case, we write

$$\lim_{n \rightarrow \infty} x_n = +\infty.$$

Definition 18. Let (x_n) be a real sequence. We say that (x_n) diverges to $-\infty$ if, for every $R \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies x_n < R.$$

In this case, we write

$$\lim_{n \rightarrow \infty} x_n = -\infty.$$

Example 28. The sequence (n^2) diverges to $+\infty$, whereas the sequence $(-n)$ diverges to $-\infty$. That is,

$$\lim_{n \rightarrow \infty} n^2 = +\infty \quad \text{and} \quad \lim_{n \rightarrow \infty} (-n) = -\infty.$$

Example 29. Define a sequence (x_n) by

$$x_{2k-1} = 1, \quad x_{2k} = 2k, \quad k \in \mathbb{N}.$$

The sequence (x_n) is unbounded but does not diverge to $+\infty$.

Indeed, let $R = 1$. For any $N \in \mathbb{N}$, choose $n = 2N - 1$. Then $n \geq N$, but

$$x_n = x_{2N-1} = 1 \not> R.$$

Hence (x_n) does not diverge to $+\infty$.

Example 30. Let $a > 1$. Then

$$\lim_{n \rightarrow \infty} a^n = +\infty.$$

Indeed, write $a = 1 + h$ for some $h > 0$. By Bernoulli's inequality,

$$a^n = (1 + h)^n \geq 1 + nh > nh.$$

Let $R \in \mathbb{R}$ be given. By the Archimedean property, there exists $N \in \mathbb{N}$ such that

$$Nh > R.$$

Then, for every $n \geq N$,

$$a^n > nh \geq Nh > R.$$

Hence $a^n \rightarrow +\infty$.

Example 31. The sequence $((n!)^{1/n})$ diverges to $+\infty$.

Let $R > 0$ be given. Since

$$\lim_{n \rightarrow \infty} \frac{R^n}{n!} = 0,$$

there exists $N \in \mathbb{N}$ such that

$$\frac{R^n}{n!} < 1, \quad n \geq N.$$

Thus

$$R^n < n!,$$

and therefore

$$R < (n!)^{1/n}, \quad n \geq N.$$

Hence

$$(n!)^{1/n} \rightarrow +\infty.$$

Example 32. Consider the sequence

$$x_n = (-1)^n.$$

This sequence is divergent. Moreover, it diverges neither to $+\infty$ nor to $-\infty$.

Indeed, the sequence is bounded since $-1 \leq x_n \leq 1$ for all n . Hence, it cannot diverge to $+\infty$ or to $-\infty$. On the other hand, we have already shown that (x_n) is not convergent. Therefore, (x_n) is divergent, but not divergent to $\pm\infty$.

Example 33. Define

$$x_n := \sum_{k=1}^n \frac{1}{k}, \quad n \in \mathbb{N}.$$

We show that (x_n) diverges to $+\infty$.

Let $R \in \mathbb{R}$ be given. By the Archimedean Property, choose $m \in \mathbb{N}$ such that

$$\frac{m}{2} > R.$$

Now let $N := 2^m$. For every $n \geq N$, we have

$$x_n \geq x_{2^m}.$$

Moreover,

$$\begin{aligned} x_{2^m} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \cdots + \frac{1}{8}\right) + \cdots \\ &\quad + \left(\frac{1}{2^{m-1} + 1} + \cdots + \frac{1}{2^m}\right). \end{aligned}$$

Each group after the first two contains at least

$$2^{j-1} \cdot \frac{1}{2^j} = \frac{1}{2},$$

where $j = 1, 2, \dots, m$. Hence

$$x_{2^m} \geq 1 + \frac{m}{2} > R.$$

Therefore,

$$x_n \geq x_{2^m} > R, \quad n \geq N.$$

This proves that

$$\lim_{n \rightarrow \infty} x_n = +\infty.$$

Exercise 14. Let

$$x_n := 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}.$$

1. Show that (x_n) is monotonically increasing.
2. Show that (x_n) is bounded above.
3. Deduce that (x_n) is convergent.
4. Show that

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

2.7 Ratio and root tests

Theorem 13 (Ratio Test). *Let (u_n) be a sequence of positive real numbers such that*

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \ell.$$

Then the following holds.

1. *If $0 \leq \ell < 1$, then*

$$\lim_{n \rightarrow \infty} u_n = 0.$$

2. *If $\ell > 1$, then*

$$\lim_{n \rightarrow \infty} u_n = +\infty.$$

Remark 3. *If $\ell = 1$, no conclusion can be drawn.*

For example,

$$u_n = n$$

satisfies

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{n+1}{n} = 1,$$

but (u_n) diverges to $+\infty$.

On the other hand,

$$u_n = \frac{n+1}{n}$$

also satisfies

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = 1,$$

but

$$\lim_{n \rightarrow \infty} u_n = 1.$$

Theorem 14 (Root Test). *Let (u_n) be a sequence of positive real numbers such that*

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \ell.$$

Then the following holds.

1. *If $0 \leq \ell < 1$, then*

$$\lim_{n \rightarrow \infty} u_n = 0.$$

2. *If $\ell > 1$, then*

$$\lim_{n \rightarrow \infty} u_n = +\infty.$$

Remark 4. If $\ell = 1$, no conclusion can be drawn.

For example,

$$u_n = n$$

satisfies

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = 1,$$

but (u_n) diverges to $+\infty$.

On the other hand,

$$u_n = \frac{1}{n}$$

also satisfies

$$\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = 1,$$

but

$$\lim_{n \rightarrow \infty} u_n = 0.$$

Example 34. Show that

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0.$$

Indeed,

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{2^{n+1}/(n+1)!}{2^n/n!} = \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0.$$

Hence, the conclusion follows from the Ratio Test.

Example 35. Show that

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^n = 0.$$

Indeed,

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{1}{2}\right)^n} = \frac{1}{2} < 1.$$

Hence, the conclusion follows from the Root Test.

2.8 Subsequences

Definition 19. Let (r_n) be a strictly increasing sequence of natural numbers that is

$$r_{n+1} > r_n \quad \text{and} \quad r_n \geq n \quad \text{for every} \quad n \in \mathbb{N}.$$

If (x_n) is any sequence, then the sequence

$$(x_{r_n})$$

is called a subsequence of (x_n) .

One can view a subsequence of (x_n) as an infinite part of the sequence obtained by deleting some terms, while keeping the remaining terms in the same order.

Example 36. Let

$$x_n = 2^n, \quad n \in \mathbb{N}.$$

Then

$$\{2^{2n}\}_{n \in \mathbb{N}} = \{4, 16, 64, \dots\}$$

is a subsequence of (x_n) .

Here, the corresponding indices are

$$r_n = 2n, \quad n \in \mathbb{N},$$

which form a strictly increasing sequence.

Remark 5. A subsequence of (x_n) can also be written as $(x_{f(n)})$, where

$$f : \mathbb{N} \rightarrow \mathbb{N}$$

is a strictly increasing function. Here, the n th term of the subsequence is the $f(n)$ th term of the original sequence.

Example 37. Let (x_n) be any sequence.

1. The sequence

$$(x_2, x_4, x_6, \dots)$$

is called the even subsequence of (x_n) . Here,

$$r_n = 2n, \quad n \in \mathbb{N}.$$

2. The sequence

$$(x_1, x_3, x_5, \dots)$$

is called the odd subsequence of (x_n) . Here,

$$r_n = 2n - 1, \quad n \in \mathbb{N}.$$

Theorem 15. Let (x_n) be a sequence converging to x . If (x_{r_n}) is a subsequence of (x_n) , then

$$x_{r_n} \rightarrow x.$$

Proof. Let $\varepsilon > 0$ be given. Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

Since (r_n) is a strictly increasing sequence of natural numbers, we have $r_n \geq n$ for every $n \in \mathbb{N}$. Hence, if $n \geq N$, then

$$r_n \geq n \geq N,$$

and therefore

$$|x_{r_n} - x| < \varepsilon.$$

Thus,

$$x_{r_n} \rightarrow x.$$

□

Exercise 15. Let (x_n) be a sequence. Suppose that both the even subsequence (x_{2n}) and the odd subsequence (x_{2n-1}) converge to the same limit x . Show that the original sequence (x_n) also converges to x .

Example 38. Show that

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{2n}\right)^n = \sqrt{e}.$$

Solution. Let

$$x_n = \left(1 + \frac{1}{n}\right)^n.$$

We have already shown that $x_n \rightarrow e$. Hence, every subsequence of (x_n) also converges to e . In particular,

$$x_{2n} = \left(1 + \frac{1}{2n}\right)^{2n} \rightarrow e.$$

Therefore,

$$\left[\left(1 + \frac{1}{2n}\right)^n\right]^2 \rightarrow e.$$

Since $\left(1 + \frac{1}{2n}\right)^n > 0$ for every n , it follows that

$$\left(1 + \frac{1}{2n}\right)^n \rightarrow \sqrt{e}.$$

Theorem 16 (Existence of a Monotone Subsequence). *Every real sequence has a monotone subsequence.*

Theorem 17 (Bolzano–Weierstrass Theorem). *Every bounded real sequence has a convergent subsequence.*

The proofs of the above two theorems are beyond the scope of this course and are therefore omitted. Interested students may consult a standard text on Real Analysis for their proofs.

Example 39. *Assume that the sequence $(n^{1/n})$ is convergent. Show that*

$$\lim_{n \rightarrow \infty} n^{1/n} = 1.$$

Solution. Let

$$n^{1/n} \longrightarrow \ell.$$

Since $((2n)^{1/(2n)})$ is a subsequence of $(n^{1/n})$, we also have

$$(2n)^{1/(2n)} \longrightarrow \ell.$$

Hence

$$((2n)^{1/(2n)})^2 \longrightarrow \ell^2.$$

On the other hand,

$$((2n)^{1/(2n)})^2 = 2^{1/n} n^{1/n}.$$

Since

$$2^{1/n} \longrightarrow 1,$$

it follows from the algebra of limits that

$$2^{1/n} n^{1/n} \longrightarrow \ell.$$

Therefore,

$$\ell^2 = \ell.$$

Since $n^{1/n} \geq 1$ for every n , we have $\ell \geq 1$. Thus,

$$\ell = 1.$$

Example 40. *Show that the sequence $((-1)^n)$ is divergent.*

Solution. Suppose, on the contrary, that $((-1)^n)$ converges to some real number ℓ . Then every subsequence must also converge to ℓ .

However, the even subsequence is

$$((-1)^{2n}) = (1, 1, 1, \dots),$$

which converges to 1, whereas the odd subsequence is

$$((-1)^{2n-1}) = (-1, -1, -1, \dots),$$

which converges to -1 .

Since a convergent sequence cannot have two subsequences with different limits, we obtain a contradiction. Hence

$$((-1)^n)$$

is divergent.

Exercise 16. Apply the Bolzano–Weierstrass Theorem to solve the following problems:

1. Show that the sequence $(\sin n)$ has a convergent subsequence.
2. Prove that every bounded sequence has a monotone convergent subsequence.
3. True or False: A sequence (x_n) is bounded if and only if every subsequence of (x_n) has a convergent subsequence.

3 Infinite series

In the previous chapter, we studied sequences, that is, ordered infinite collections of real numbers. A natural question now arises:

Can we meaningfully add infinitely many real numbers?

For any finite collection of real numbers

$$x_1, x_2, \dots, x_n,$$

the sum

$$x_1 + x_2 + \dots + x_n$$

is well defined. It is therefore tempting to define the “sum” of an infinite sequence

$$(x_n)$$

by writing

$$\sum_{n=1}^{\infty} x_n.$$

However, unlike finite sums, infinite sums require much greater care. If we manipulate infinitely many terms without a precise definition, we may arrive at contradictory conclusions.

For example, consider the sequence

$$x_n = (-1)^{n-1},$$

so that

$$1 - 1 + 1 - 1 + \cdots .$$

If we group the terms as

$$(1 - 1) + (1 - 1) + (1 - 1) + \cdots ,$$

then we obtain

$$0 + 0 + 0 + \cdots = 0.$$

On the other hand, if we write

$$1 + (-1 + 1) + (-1 + 1) + \cdots ,$$

then we obtain

$$1 + 0 + 0 + \cdots = 1.$$

Thus, the same infinite expression appears to have two different values:

$$1 - 1 + 1 - 1 + \cdots = 0 \quad \text{and} \quad 1 - 1 + 1 - 1 + \cdots = 1.$$

This contradiction shows that an infinite sum cannot be treated in the same way as a finite sum. We therefore need a rigorous definition of what it means for an infinite series to have a sum. This is the main objective of this chapter.

3.1 Convergence and sum of an infinite series

Definition 20. Let (a_n) be a sequence of real numbers. A formal expression of the form

$$\sum_{n=1}^{\infty} a_n,$$

or simply

$$\sum a_n \quad \text{or} \quad \sum_n a_n$$

is called an infinite series.

Definition 21 (Partial Sum). *Let*

$$\sum_{n=1}^{\infty} a_n$$

be an infinite series. For each $n \in \mathbb{N}$, the finite sum

$$s_n := a_1 + a_2 + \cdots + a_n$$

is called the n th partial sum of the series.

Notation. Associated with the series

$$\sum_{n=1}^{\infty} a_n,$$

we have the sequence of partial sums

$$(s_n),$$

where

$$s_n = a_1 + a_2 + \cdots + a_n, \quad n \in \mathbb{N}.$$

Thus, every infinite series naturally determines a sequence of partial sums.

Definition 22 (Convergent Series). *An infinite series*

$$\sum_{n=1}^{\infty} a_n$$

is said to be convergent if the sequence of partial sums (s_n) is convergent.

Notation. If

$$s_n \longrightarrow s,$$

then we call s the *sum* of the series and write

$$\sum_{n=1}^{\infty} a_n = s.$$

Definition 23 (Divergent Series). *An infinite series*

$$\sum_{n=1}^{\infty} a_n$$

is said to be divergent if the sequence of partial sums (s_n) is divergent.